Scalar local linearization

Linear approximation: for scalar-valued f whose first partial derivatives exist at $\mathbf{a} \in \mathbb{R}^n$,

$$\begin{aligned} L(\mathbf{x}) &= f(\mathbf{a}) + \nabla f(\mathbf{a})^T(\mathbf{x} - \mathbf{a}) \\ &= f(\mathbf{a}) + \sum_{i=1}^{n} f_{x_i}(\mathbf{a})(x_i - a_i). \end{aligned}$$

This is **affine** in $\mathbf{x}$. Its **linear part** is $\mathbf{h} \mapsto \nabla f(\mathbf{a}) \cdot \mathbf{h}$ on input changes. For differentiable f and small $\mathbf{h}$, $f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) \approx \nabla f(\mathbf{a}) \cdot \mathbf{h}$.

Two variables: $L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$.

Differentiability

Definition: f is differentiable at $\mathbf{a}$ if its gradient exists there and

$$\lim_{\mathbf{x} \to \mathbf{a}} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|} = 0.$$

The approximation error is smaller than first order in the input change. Existence of partial derivatives alone does not ensure this limit.

Sufficient condition: if $\mathbf{a}$ is an interior point of the domains of f and all its first partials, and each partial is continuous at $\mathbf{a}$, then f is differentiable at $\mathbf{a}$.

Vector-valued differentiability: F is differentiable at $\mathbf{a}$ if and only if each component F_i is differentiable there.

Tangent planes and level curves

Tangent plane: a plane through a point of a surface that contains the tangent lines of differentiable curves on the surface through that point.

Graph theorem: if $f(x, y)$ is differentiable at (a, b), its graph has a unique tangent plane at $(a, b, f(a, b))$:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

It is the graph $z = L(x, y)$.

Gradient and level curves: if $f(x, y)$ has continuous first partials near (a, b) and $\nabla f(a, b) \neq \mathbf{0}$, then $\nabla f(a, b)$ is perpendicular to the level curve through (a, b).

Vector-valued local linearization

Linear approximation: for $F : \mathbb{R}^n \to \mathbb{R}^m$ with all first component partials at $\mathbf{a}$,

$$L(\mathbf{x}) = F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x} - \mathbf{a}), \qquad L_i(\mathbf{x}) = F_i(\mathbf{a}) + \nabla F_i(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}).$$

Here $J_F(\mathbf{a})$ is $m \times n$. The approximation is affine in $\mathbf{x}$; $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ is its linear part.

Local prediction: if F is differentiable at $\mathbf{a}$, then $F(\mathbf{a} + \mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}$ for small $\mathbf{h}$. Each component uses its scalar linear approximation.